

Late 1950's

1

WU $\rightarrow$ P violated in weak interactions
 $\uparrow$ ONLY

θ/τ PUZZLE: $\begin{cases} \text{"}\theta\text{"} \equiv K^+ \rightarrow (2\pi) \leftarrow P=+1 \\ \text{"}\tau\text{"} \equiv K^+ \rightarrow (3\pi) \leftarrow P=-1 \end{cases}$

~~the particles can share mass~~

same mass and lifetime (within exp. resolution): is it same particle?

How can it decay to different parity?

it's ok: it's a weak decay

($\Rightarrow$ can violate P)

But there's another problem: $K^\pm$ decays weakly.

But it's produced strong -- Why doesn't it decay strong?

$$\tau_{\text{strong}} \sim 10^{-22} \text{ s} \quad \tau_{\text{weak}} \sim 10^{-10} \text{ s}$$

It's "strange" $\rightarrow$ NEW QUANTUM NUMBER: STRANGENESS
CONSERVED BY STRONG INT, VIOLATED BY WEAK INT

$$K^+: S = +1$$

$$K^-: S = -1$$

$K^\pm \xrightarrow{\text{strong}} \text{hadrons}$

because it's the lightest particle with $S \neq 0$

$$\text{eg } \underset{S=1}{K^+} \rightarrow \underset{S=0}{\pi^+ \pi^+ \pi^-} \quad \Delta S = 1$$

What about neutral kaons?

2

	K^0	$\bar{K}^0$
Q	0	0
B	0	0
S	+1	-1

FIRST NEUTRAL MESON
WITH (A) CHARGE

FIRST NEUTRAL BOSON
FOR WHICH
PARTICLE $\neq$ ANTIPARTICLE

eg: γ, π^0

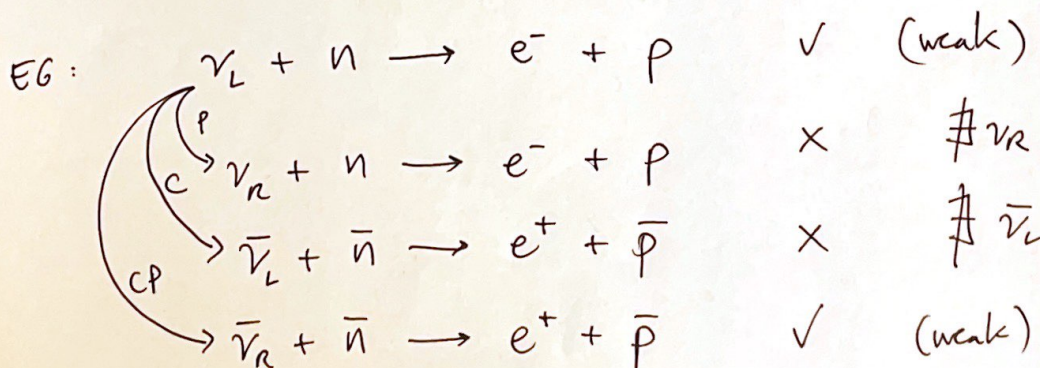
What is the nature of $K^0/\bar{K}^0$?

they differ only by S , a quantum number not conserved by weak interactions. Does this mean that from p.o.v. of weak int. they are same state?

By now P had fallen, but people thought that

(CP) was a good symmetry of the Universe

 red symmetry between matter and anti-matter



After fall of P , CP was last button standing

Let's look at $K^0/\bar{K}^0$

3

Produced strongly $\rightarrow$ eigenstates of strong interaction

But they are NOT eigenstates of CP

$$CP |K^0\rangle = |\bar{K}^0\rangle \quad \begin{cases} P|K^0\rangle = -|K^0\rangle \\ C|K^0\rangle = e^{i\alpha} |\bar{K}^0\rangle \end{cases}$$

(IF) CP is conserved $\Rightarrow$ the physical states are the eigenstates of CP and α so that $CP|K^0\rangle = |\bar{K}^0\rangle$

There are:

$$|K_1^0\rangle = \frac{1}{\sqrt{2}} (|K^0\rangle + |\bar{K}^0\rangle) \quad CP = +1$$

$$|K_2^0\rangle = \frac{1}{\sqrt{2}} (|K^0\rangle - |\bar{K}^0\rangle) \quad CP = -1$$

We know that both K^0 and $\bar{K}^0$ can decay to 2π

$$K^0 \rightarrow \pi^+ \pi^- \quad \bar{K}^0 \rightarrow \pi^+ \pi^-$$

The $(\pi^+ \pi^-)$ state has $CP = +1$

In fact 

In fact both C and P switch the pions

4

$$\pi^- \leftrightarrow \pi^+$$

P is a symmetry of the wavefunction

$$C \text{ does } C(\pi^\pm) = \overline{\pi^\pm} = \pi^\mp$$

$\Rightarrow$ depends on symmetry of wavefunction

BUT $L=0$ because K have $J=0$
(lightest)

$$\Rightarrow P = (-1)^{L=0} = +1$$

$$C = (-1)^{L=0} = +1$$

And then there's the ^{INTRINSIC} parity of the pions

$$P(\pi^+) = P(\pi^-) = -1$$

$$\begin{aligned} \Rightarrow CP(\pi^+\pi^-) &= (CP)_{\text{spatial}} \cdot P(\pi^+) \cdot P(\pi^-) = \\ &= \overset{C}{(+1)} \cdot \overset{P}{(+1)} \cdot (-1) \cdot (-1) = +1 \end{aligned}$$

So if CP is conserved ONLY $|K_1^0\rangle$
can decay to 2π

[5]

$$K_1^0 \rightarrow 2\pi \quad K_2^0 \not\rightarrow 2\pi$$

Using similar arguments $CP(3\pi) = -1$

$$\Rightarrow K_1^0 \not\rightarrow 3\pi \quad K_2^0 \rightarrow 3\pi$$

$$m_\pi \sim 140 \text{ MeV} \quad m_K \sim 500 \text{ MeV}$$

$$\Rightarrow m(3\pi) \lesssim m_K$$

phase space available to 2π decay MUCH larger

$$\Rightarrow \text{expect } \tau(K_1^0) \ll \tau(K_2^0)$$

This is why we call them

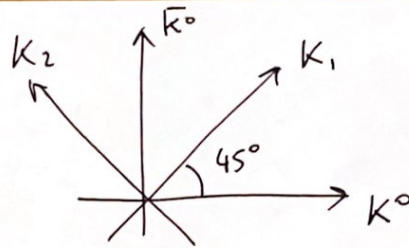
$$K_1^0 = K_S^0 \quad \text{"short"} \quad \tau_S \sim 0.9 \cdot 10^{-10} \text{ s}$$

$$K_2^0 = K_L^0 \quad \text{"long"} \quad \tau_L \sim 0.5 \cdot 10^{-7} \text{ s}$$

$\uparrow$
 $\times 10^3$

RECAP

two representations



6

STRONG FORCE EIGENSTATES

PRODUCTION

$|K^0\rangle, |\bar{K}^0\rangle$

CP EIGENSTATES

TIME EVOLUTION

$|K_1^0\rangle, |K_2^0\rangle$

$$|K^0\rangle = \frac{1}{\sqrt{2}} (|K_1^0\rangle + |K_2^0\rangle)$$

$$|\bar{K}^0\rangle = \frac{1}{\sqrt{2}} (|K_1^0\rangle - |K_2^0\rangle)$$

$$|K_{1,2}^0\rangle = \frac{1}{\sqrt{2}} (|K^0\rangle \pm |\bar{K}^0\rangle)$$

↑
if CP conserved these are
Hamiltonian eigenstates too

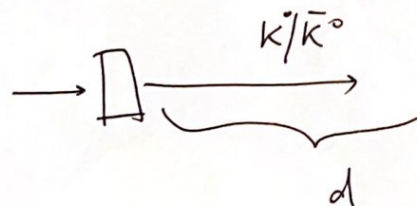
So if we have an experiment



↑
KAONS produced in strong interaction

⇒ at $t=0$ $|K^0\rangle$ and $|\bar{K}^0\rangle$

But if they then travel some distance d



time evolution depends
on H eigenstates



$$|K_1^0(t)\rangle = e^{-im_1 t - \Gamma_1 t/2} \frac{1}{\sqrt{2}} \left[|K_0(0)\rangle + |\bar{K}^0(0)\rangle \right] \quad \boxed{7}$$

$$|K_2^0(t)\rangle = e^{-im_2 t - \Gamma_2 t/2} \frac{1}{\sqrt{2}} \left[|K_0(0)\rangle - |\bar{K}^0(0)\rangle \right]$$

FIRST CONSEQUENCE : OSCILLATIONS

If a state is produced at $t=0$ as purely K^0 when it propagates it will oscillate

$$K^0 \rightarrow \bar{K}^0 \rightarrow K^0$$

with amplitudes

$$\langle K^0 | \psi(t) \rangle = \frac{1}{2} \left(e^{-im_1 t - \Gamma_1 t/2} + e^{-im_2 t - \Gamma_2 t/2} \right)$$

$$\langle \bar{K}^0 | \psi(t) \rangle = \frac{1}{2} \left(e^{-im_1 t - \Gamma_1 t/2} - e^{-im_2 t - \Gamma_2 t/2} \right)$$

SECOND CONSEQUENCE: REGENERATION

8

Start at $t=0$ with pure K^0 beam in vacuum

50% K_1^0

50% K_2^0

REMEMBER: $\tau(K_1^0) \sim 10^{-10} \text{ s}$ K_1 short

$\tau(K_2^0) \sim 10^{-7} \text{ s}$ K_2 long

$\Rightarrow$ if you wait $10 \tau_1 \rightarrow$ all K_1^0 decay

$\Rightarrow$ left with only K_2^0

at $t=0$ 100% $K^0 \Leftrightarrow 50\% K_1^0 / 50\% K_2^0$

at $t = 10 \tau_1 \sim 100\% K_2^0 \Leftrightarrow 50\% K^0 / 50\% \bar{K}^0$

Now that we have pure K_2^0 beam we make it pass through matter $\Rightarrow$ strong interaction

$\nearrow$
 K^0 and $\bar{K}^0$ are the eigenstates

But K^0 and $\bar{K}^0$ have different interaction with matter 9

$$K^0 + p \rightarrow K^0 + p$$

$$K^0 + n \rightarrow K^0 + n$$

$$K^0 + p \rightarrow K^+ + n$$

$$\bar{K}^0 + p \rightarrow \bar{K}^0 + p$$

$$\bar{K}^0 + n \rightarrow \bar{K}^0 + n$$

$$\bar{K}^0 + p \rightarrow \pi^+ + \Lambda^0$$

$$\bar{K}^0 + p \rightarrow \pi^+ + \Sigma^+$$

$$\bar{K}^0 + n \rightarrow \pi^0 + \Lambda^0$$

MORE CHANNELS AVAILABLE TO $\bar{K}^0$!

this is because $|\bar{K}^0\rangle = |\bar{d}s\rangle$

↑ quark can swap with one of the quarks in p/n

eg. $|\bar{K}^0\rangle = |\bar{d}s\rangle$

$|p\rangle = |uud\rangle$

$|u\bar{d}\rangle = |\pi^+\rangle$

$|uds\rangle = |\Lambda^0\rangle$

$|K^0\rangle = |d\bar{s}\rangle$

↑ this is an antiquark so not so easy

$\Rightarrow |\bar{K}^0\rangle$ interacts more in matter

$\Rightarrow |\bar{K}^0\rangle$ component more strongly absorbed!

$\Rightarrow$ if you start with 50% K^0 / 50% $\bar{K}^0$ 10

$\Rightarrow$ you end up with less $\bar{K}^0$ w/ K^0

BUT so if at $t=t'$ $|\psi(t')\rangle = 100\% |K_2^0\rangle$
50% K^0 / 50% $\bar{K}^0$

then at $t > t'$ (in matter) you get a different
 $K^0/\bar{K}^0$ mix

$\Rightarrow K_1^0$ regeneration

OK BACK TO ORIGINAL QUESTION: if CP is conserved

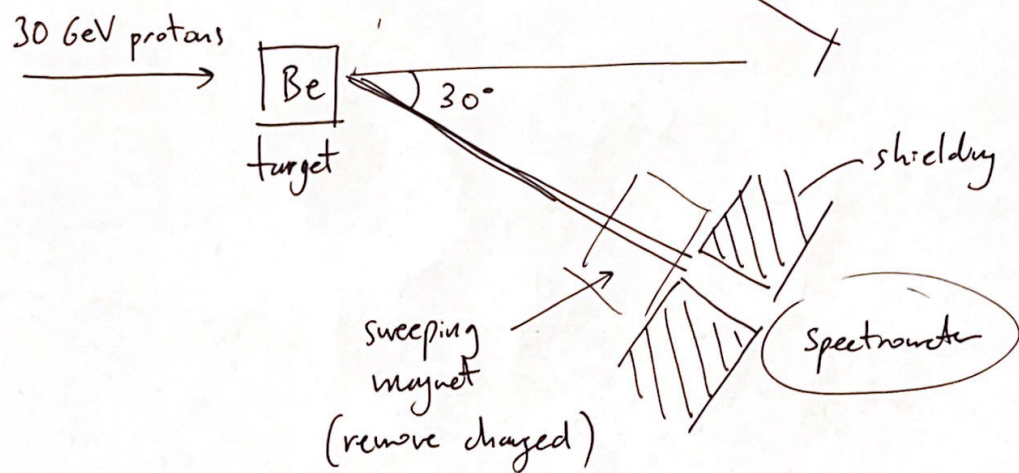
$\Rightarrow K_1^0$ and K_2^0 are eigenstates of H

$\Rightarrow K_1^0 \rightarrow \pi^+\pi^-$ (2π) CP = +1

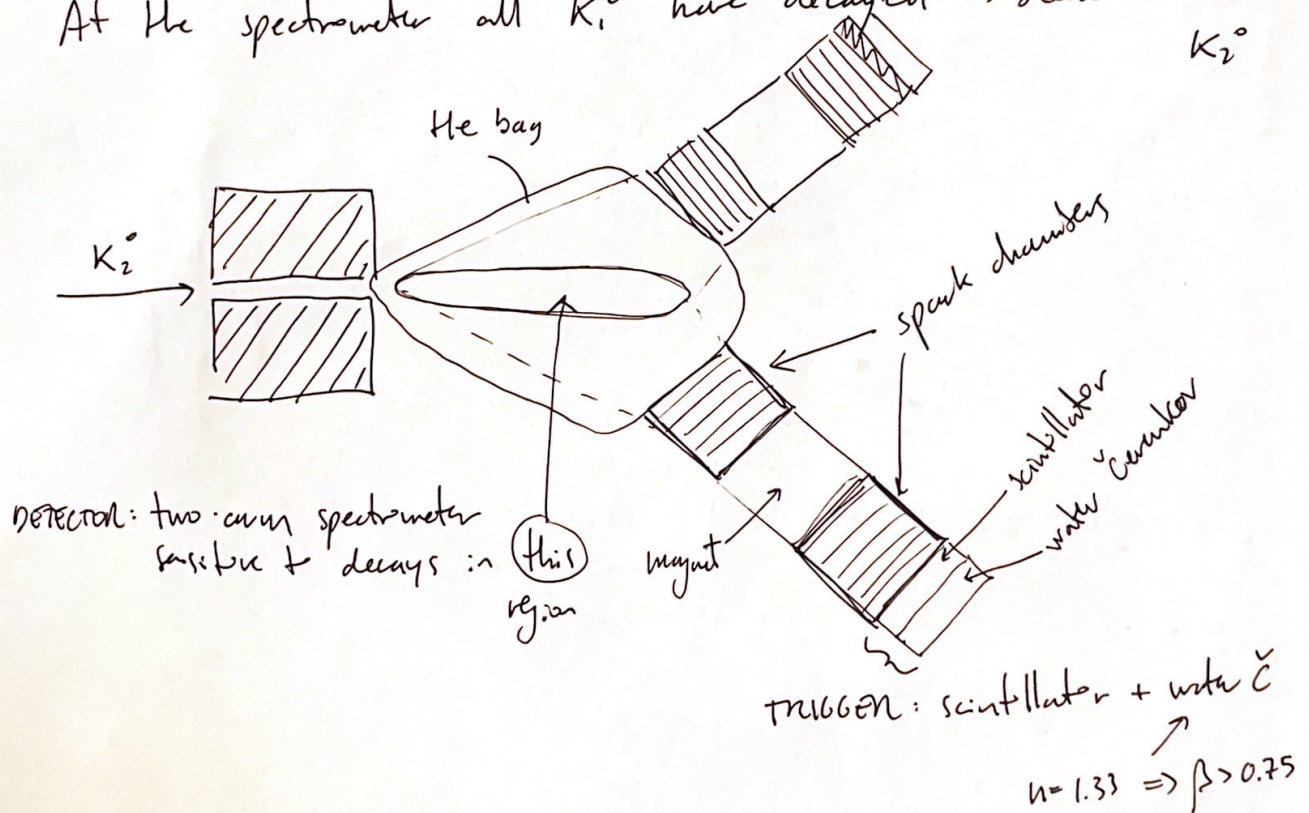
$K_2^0 \rightarrow 3\pi$ CP = -1

IF CP GOOD SYMMETRY $K_2^0 \not\rightarrow 2\pi$ (even)

1964 Cronin & Fitch 17 m $\left(ct_1 \sim 3 \text{ cm} \right. \left. ct_2 \sim 15 \text{ m} \right)$ 11



At the spectrometer all K_1^0 have decayed $\Rightarrow$ beam is 100% K_2^0

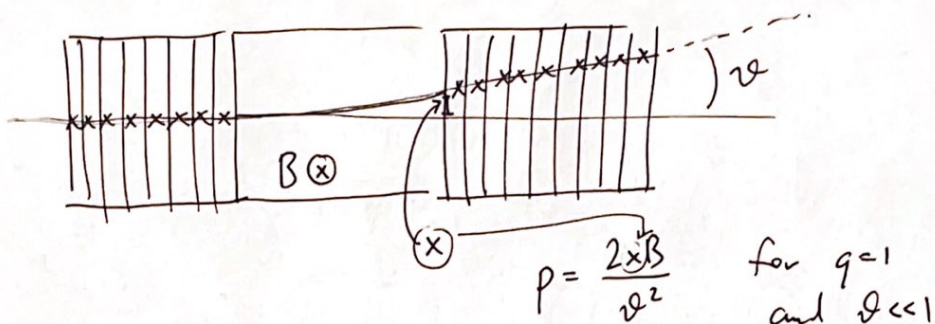


LOOKING FOR: $K_2^0 \rightarrow \pi^+ \pi^-$

CP violating decay

spark chambers + magnet $\rightarrow$ momentum measurement

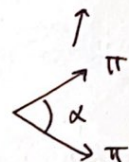
12



$$\vec{p}_1 \text{ and } \vec{p}_2 \xrightarrow{\text{assume } m_\pi} p_1, p_2$$

$$\Rightarrow M_{inv}(\pi_1, \pi_2) = |p_1 + p_2| \approx \sqrt{2E_1 E_2 (1 - \cos \alpha)}$$

$E \gg m_\pi$



(IF)

$$K_2^0 \rightarrow \pi^+ \pi^-$$

$$\Rightarrow M(\pi^+, \pi^-) = M(K_2^0) = 498 \text{ MeV}$$

and $\theta(\pi^+, \pi^-) = \theta(K_2^0) = 0$ (initial state)

θ of $\vec{p}_1 + \vec{p}_2$

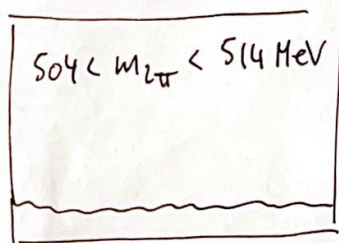
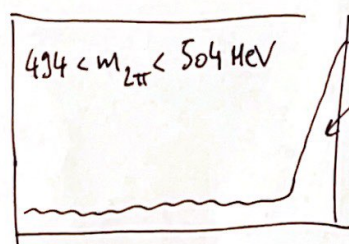
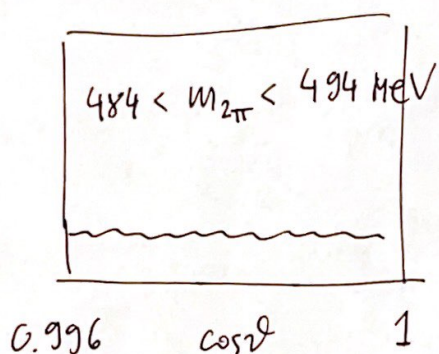
INSTEAD

$$K_2^0 \rightarrow \pi^+ \pi^- \pi^0$$

∇ does not peak!
(missing a piece!)

RESULTS:

13

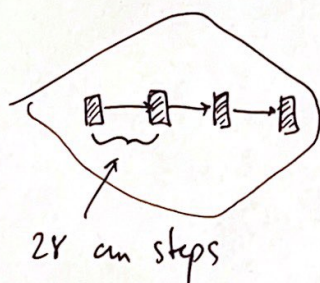


! OBSERVATION OF $K_L^0 \rightarrow \pi^+ \pi^-$?

Could it be something else?

What about K_S^0 regeneration in the He bag?

↑
To study this possibility, added tungsten targets inside the bag



Tungsten is dense $\rightarrow$ more interaction with matter
 $\rightarrow$ more regeneration

K_S^0 regeneration is observed in tungsten in accordance with expectations (calculations)

$\Rightarrow$ same calculations applied to He-only bag say that contribution of K_S^0 regeneration should be $\sim 10^4$ times lower than observed even

$K_L^0 \rightarrow 3\pi$ doesn't give a peak
and neither do other K_L^0 decays

$$K_L^0 \rightarrow \pi^\pm e^\mp \nu \quad K_L^0 \rightarrow \pi^\pm \mu^\mp \nu$$

mistakenly taking charged lepton for π

Finally $K_L^0 \rightarrow \pi^\pm \pi^\mp \gamma$ is possible, but to contribute
needs $E_\gamma < 1 \text{ MeV} \rightarrow$ too rare

$$\Rightarrow \boxed{K_L^0 \rightarrow \pi^+ \pi^-} ! \Rightarrow \text{CP violation!}$$

$\Rightarrow$ CP eigenstates are NOT eigenstates of H!

Eigenstates of H / WEAK INTERACTION:

$$K_L^0 = \frac{1}{\sqrt{1+\epsilon^2}} [K_2^0 + \epsilon K_1^0]$$

$$K_S^0 = \frac{1}{\sqrt{1+\epsilon^2}} [K_1^0 + \epsilon K_2^0]$$

$(\epsilon) =$ AMOUNT OF
CP VIOLATION
IN SM

measured: $\epsilon = \frac{K_L^0 \rightarrow 2\pi}{K_L^0 \rightarrow \text{all}} \sim (2.0 \pm 0.4) \cdot 10^{-3}$
small but $\neq 0!$

STRONG EIGENSTATES:

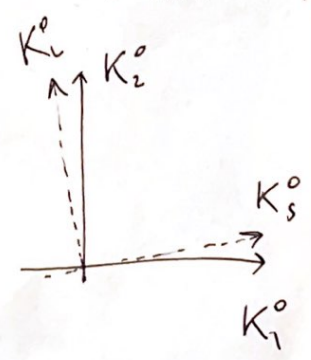
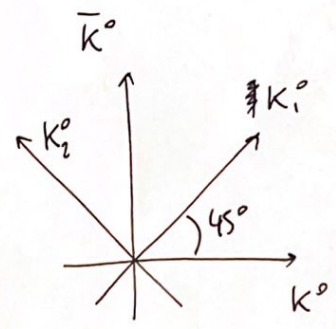
$$|K^0\rangle \quad |\bar{K}^0\rangle$$

CP EIGENSTATES:

$$|K_1^0\rangle \quad |K_2^0\rangle$$

WEAK FORCE EIGENSTATES:

$$|K_S^0\rangle \quad |K_L^0\rangle$$



A BRIEF ORITVANY

15

† 1956 PARITY (WU)

† 1964 CP (CHONIN + FITCH)

BUT CPT HOLDS ← THEOREM

⇒ IF CPT FAILS, WHOLE HOUSE BURNS DOWN

↑
(CPT)